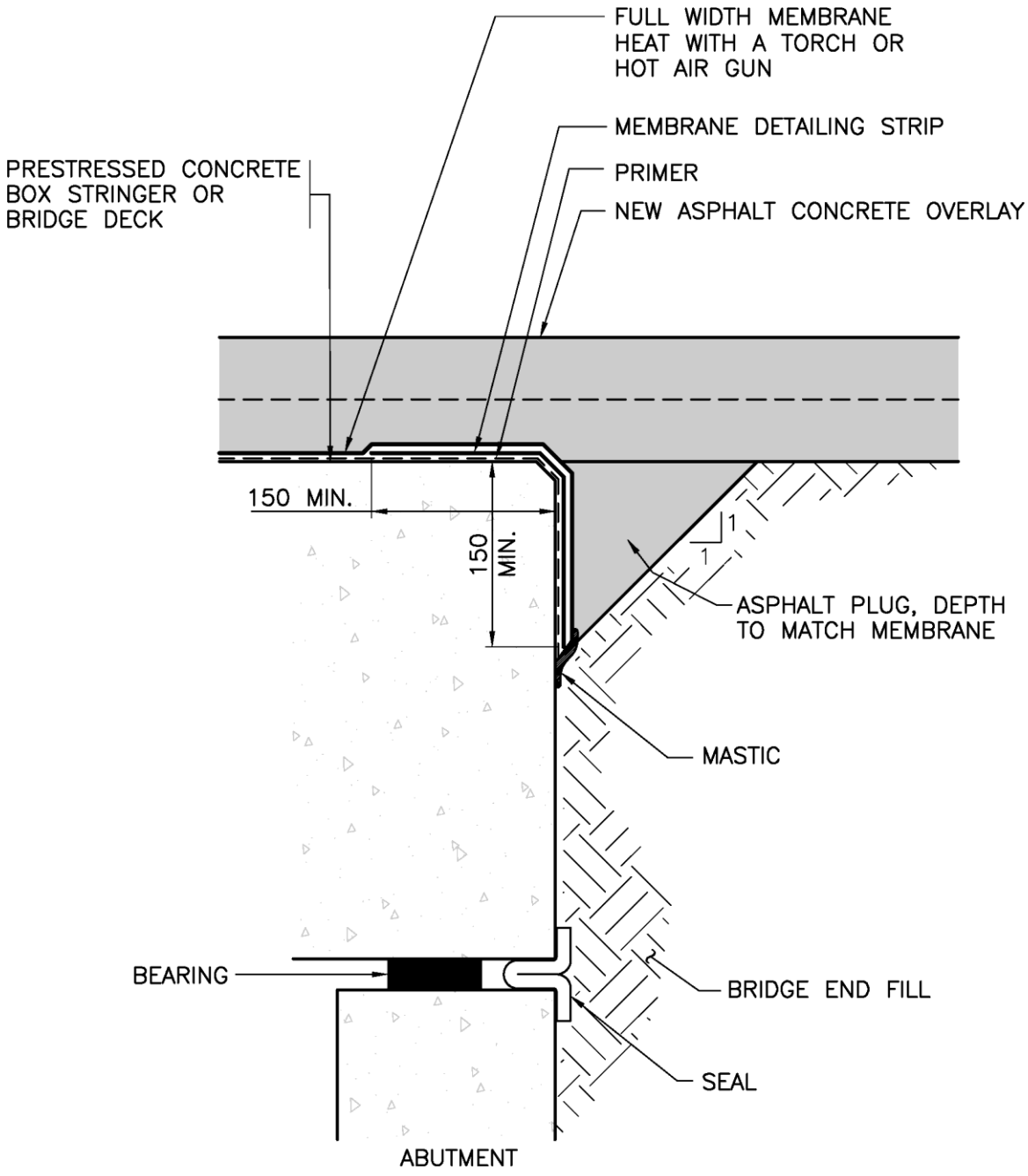
BRIDGE DECK – MEMBRANE APPLICATION
TYPICAL DETAIL WORK – BRIDGE END

SP419–07

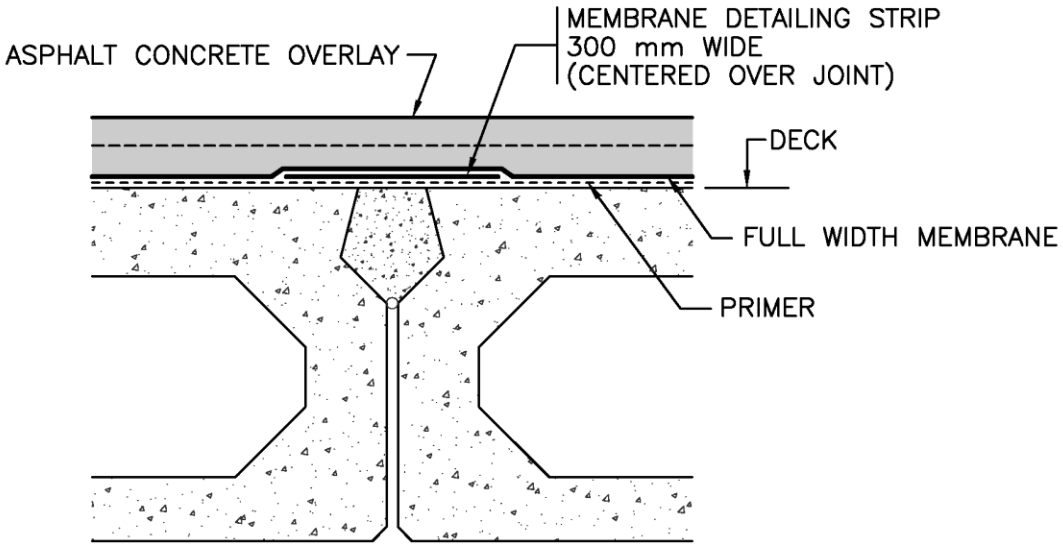


BRIDGE END

(NEW CONSTRUCTION OR EXISTING STRUCTURE WITHOUT EXISTING PAVEMENT)

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – MEMBRANE APPLICATION TYPICAL DETAIL WORK – SHEAR KEYS	SP419–08



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 420

HOT APPLIED RUBBERIZED ASPHALT WATERPROOFING MEMBRANE SYSTEM

DESCRIPTION

420.01 Scope – This specification covers the supply and installation of a Hot Applied Rubberized Asphalt Waterproofing Membrane System and an overlay of hot applied asphalt pavement.

Construction shall conform to SS Drawings SP420-01 through SP420-07 inclusive, except as modified by the Special Provisions or Drawings.

The area to be covered by the membrane system, and additional details, shall be as shown in the Drawings and Special Provisions, or as otherwise directed by the Ministry Representative.

NOTE: Waterproofing membrane systems shall not be used on Ministry concrete bridge decks contaminated with acid-soluble chloride ion (Cl^-) in excess of 0.10% by mass of cement. Membrane systems shall only be used on bridges that have been designed to accommodate the thickness and dead load of the system.

420.02 Referenced Standards – The ASTM and Ontario Provincial Standard Specification (OPSS.Prov) standards listed in Table 420-A, in the edition current at the time of Tender, shall apply unless specified otherwise herein or in the Special Provisions or Drawings.

420.03 Applicator's Qualification – Only applicator(s) trained and certified by the membrane system supplier/manufacturer whose product is used in the Work, will be permitted to install the waterproofing membrane system. Certification documentation shall be provided to the Ministry Representative upon request.

Special Provisions may identify alternate acceptable documentation for remote locations and special requirements.

420.04 Quality Control – Installation and quality of the membrane system shall be inspected and corrected by the Contractor in accordance with this SS 420 and the supplier/manufacturer's requirements. If there is a discrepancy between the two, the more stringent requirement shall apply. In cases where it may be unclear as to which requirement is more stringent, the Ministry Representative will have the final say.

The Contractor shall provide a copy of the supplier/manufacturer's quality control certification for each lot of material prior to its use in the Work.

420.05 Quality Assurance – The Ministry will implement a quality assurance program by auditing the Contractor's quality control program and by inspecting and testing at its discretion.

Table 420-A: Referenced Standards

Specification	Description
ASTM D1785	Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120
ASTM D2466	Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40
ASTM D2564	Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
ASTM D4833	Standard Test Method for Index Puncture Resistance of Geomembranes and Related Products
ASTM D6134	Standard Specification for Vulcanized Rubber Sheets Used in Waterproofing Systems
OPSS.Prov 914	Waterproofing Bridge Decks with Hot Applied Asphalt Membrane
OPSS.Prov 1212	Hot Poured Rubberized Asphalt Joint Sealing Compound
OPSS.Prov 1213	Hot-Applied Rubberized Asphalt Waterproofing Membrane
OPSS.Prov 1215	Material Specification for Protection Board
OPSS.Prov 1217	Reinforcement Used With Hot-Applied Rubberized Asphalt Waterproofing Membrane

Inspection by the Ministry shall not relieve the Contractor from obligation to perform the Work in accordance with the Contract.

MATERIALS

420.10 General

420.10.01 Membrane System – The membrane system shall be in accordance with one of the following:

(a) One of the systems from the Ministry Recognized Products List (RPL) under the category "Hot

NEW SECTION

Rubberized Asphalt Membrane System for Concrete Decks", or

(b) A system meeting the Ontario Provincial Standard Specification OPSS.Prov 914.

420.10.02 Membrane System Components – The following membrane system components shall be supplied by the supplier/manufacturer of the membrane as a complete package:

(a) primer,

(b) asphalt membrane,

(c) membrane reinforcing fabric,

(d) rubber membrane,

(e) protection board, and

(f) wick drains.

Two weeks prior to the application of the membrane system, the Contractor shall provide the Ministry Representative with a list of the supplier/manufacturer's materials for the selected membrane system and also provide detailed specifications, installation procedures, technical data sheets, safety data sheets and quality control procedures and certification documents.

420.11 System Materials

420.11.01 Primer – The primer used in conjunction with the asphalt membrane shall be in accordance with the membrane supplier/manufacturer's requirements and shall be compatible with the asphalt membrane. Primer shall be in accordance with the Ontario Provincial Standard Specification OPSS.Prov 1216.

420.11.02 Asphalt Membrane – Asphalt membrane shall be hot applied rubberized asphalt membrane and shall be supplied in cakes ready for melting and application. Asphalt membrane shall meet the Ontario Provincial Standard Specification OPSS.Prov 1213.

420.11.03 Membrane Reinforcing Fabric – Membrane reinforcing fabric shall be a spun bonded sheet structure composed of 100% continuous filament polyester fibers bonded together at their crossover points. Membrane reinforcing fabric shall be supplied in minimum widths of 300 mm. The performance of the material shall be unaffected by the heat generated by the installation of the membrane system and asphalt placement. Membrane reinforcing fabric shall meet the Ontario Provincial Standard Specification OPSS.Prov 1217.

420.11.04 Rubber Membrane – Rubber membrane shall consist of 1.2 mm thick butyl and ethylene propylene diene monomer (EPDM) rubber. The membrane shall meet the requirements of ASTM D6134.

420.11.05 Waterproofing Protection Board – The protection board or sheet shall be a durable panel of 3.60 ± 0.40 mm thickness specifically designed to provide a protective cushion between the hot mix asphaltic concrete pavement and the asphalt waterproofing membrane. It shall have a water absorption property of 5% or less and shall meet the Ontario Provincial Standard Specification OPSS.Prov 1215.

420.11.06 Wick Drain – Wick drain shall be composite polypropylene with a total thickness of 3.6 mm and supplied in 100 mm widths. The puncture strength shall be 45 N measured in accordance with ASTM D4833. The performance of the material shall be unaffected by the heat generated by the installation of the membrane system and asphalt placement.

420.11.07 Weep Drain Pipes – Weep drain pipes shall be schedule 40 PVC pipe and fittings meeting the requirements of ASTM D1785 and D2466. All fittings shall be attached to the pipe by solvent welding using solvent meeting ASTM D2564 and in accordance with the manufacturer's recommendations. Weep drain pipes shall be 50 mm diameter unless otherwise shown on the Drawings or in the Special Provisions.

420.11.08 Emulsified Edge Sealant – The edge sealant product shall be selected from the Ministry Recognized Products List under the category of "Asphalt Pavement/Longitudinal Joint Sealing Systems".

420.12 Delivery and Storage – Membrane system materials shall be delivered to the Site in unopened packages and containers with the manufacturer's labels identifying the product and a batch number.

The packages and containers shall be stored in a dry, secured place, protected from weather and excessive heat. The manufacturer's instructions for handling and storage shall be followed.

EQUIPMENT

420.20 Kettle – The kettle used to heat and mix the hot-applied rubberized asphalt membrane shall be of the double boiler oil transfer type with a built-in agitator and equipped with permanently installed dial type thermometers to measure the temperature of the melted compound and the oil.

MEMBRANE CONSTRUCTION

420.30 Traffic Restrictions – All traffic, other than construction equipment directly associated with the installation of the membrane system and placement of asphalt pavement shall be prevented from using areas of the deck that have received surface preparation until paving of that area is completed.

420.31 General – The Contractor shall perform all of the operations involved in waterproofing in sequential order,

NEW SECTION

such that there are no delays between individual operations except those necessary to meet the requirements of this SS 420 and the Contract.

Waterproofing operations shall only be carried out when the air and concrete surface temperatures are 5°C or higher and are in accordance with the manufacturer's recommended temperature ranges.

All materials with overlaps shall be installed to produce a shingling effect to prevent water from becoming trapped under or against edges.

420.32 Notice of Commencement of Waterproofing Operations – The Contractor shall give the Ministry Representative forty-eight (48) hours notice prior to commencing any waterproofing operations.

420.33 Surface Preparation – Any concrete to be covered by the membrane system, including the deck concrete, parapets, curbs, sidewalks, and medians must be completely cured for at least twenty-eight (28) days prior to the application commencing.

The surfaces of the concrete to receive waterproofing shall be prepared by abrasive blasting to expose sound laitance free concrete and shall be cleaned to remove all dirt, dust, laitance, oil etc. leaving a prepared surface satisfactory for application of the primer. Protective measures shall be taken to prevent damage to or contamination of surrounding facilities and vehicles during surface preparation and cleaning.

Milled deck surfaces shall not have a profile greater than 6 mm, peak to valley.

420.34 Primer – Primer application and waterproofing shall not commence until the Ministry Representative has accepted all preparation Work.

The primer shall be applied only when the concrete is dry and clean, and when the air and concrete surface temperatures are above both the manufacturer's recommended temperature and 5°C. No precipitation shall have occurred during the previous twenty-four (24) hours or be forecast to occur within the time required for application of the primer and membrane.

Immediately prior to the application of the primer, the concrete surface shall be air blasted to remove all dust and any other foreign material. Deck drains and weep drains shall be temporarily plugged to prevent entry of the primer and asphalt membrane material.

Primer material shall be applied with equipment acceptable to the Ministry Representative which will provide uniform application at the required rate.

The application rate shall be such that the primer material will be absorbed into the concrete, resulting in a surface that is dull and black in appearance. The application of an excessive amount of primer as indicated by puddling or a shiny black surface shall be avoided.

Waterproofing equipment or material shall not be permitted on the primer until it has fully cured and is completely tack-free.

420.35 Application of Membrane Reinforcing Fabric – Membrane reinforcing fabric shall be applied as shown on the Drawings and at construction joints, shear keys, pier diaphragms, and active cracks.

Prior to the application of the asphalt membrane to the deck, a strip of hot asphalt membrane at least 5 mm thick and wide enough to extend 150 mm on either side of construction joints, shear keys, pier diaphragms, and active cracks, shall be applied to the primed concrete surface. A strip of the membrane reinforcing fabric wide enough to extend 150 mm on either side of these construction joints, shear keys, pier diaphragms, and active cracks shall be embedded within the asphalt membrane strip while it is still hot and tacky.

420.36 Application of Butyl Rubber Membrane – Butyl rubber membrane shall be applied as shown on the Drawings and along all curbs, parapets and deck drains.

A strip of hot asphalt membrane shall be applied to the height of the top of the hot mix surface course, and 150 mm onto the deck. The butyl rubber membrane shall extend 40 mm up the vertical faces, and 110 mm onto the deck surface, and overlapped a minimum of 100 mm when multiple strips are used. The butyl rubber membrane shall be applied while the asphalt membrane is still hot and tacky.

A second coat of asphalt membrane shall then be applied to fully cover the rubber membrane.

420.37 Waterproofing Around Deck Drains and Weep Drain Pipes – The asphalt membrane shall be carefully applied around the deck drains and weep drain pipes so that a positive seal is obtained. It may be necessary to temporarily plug the drains and weep drain pipes prior to waterproofing to prevent the entrance of hot membrane.

420.38 Application of Asphalt Membrane – Cakes of asphalt membrane shall be melted in the mechanically agitated heating and mixing unit specified. This unit shall keep the contents continuously agitated until the material can be drawn, free flowing and lump-free, from the mixing unit at a temperature not exceeding that recommended by the membrane manufacturer.

Membrane shall not be applied until the primer has cured completely.

The asphalt membrane shall be applied within the temperature range recommended by the manufacturer, to the clean, primed concrete deck, to form a uniform film having a thickness of 5±1 mm.

NEW SECTION

The asphalt membrane shall be applied evenly on the deck surface by using a squeegee and checked while still fluid to ensure that the applied thickness will ensure conformance to the specified final thickness requirements.

As the membrane is being installed, the surface shall be inspected for pin holes and missed areas, which shall be repaired immediately. The installation operation shall be such that discontinuities in the membrane are avoided and any joints lapped 150 mm.

The membrane shall be applied over all waterproofed joints and cracks, and shall extend up the face of curbs, parapets, and deck drains, to the height of the top of the hot mix surface course.

Moisture in the concrete may cause bubbling in the membrane. When bubbles occur, they shall be eliminated while the membrane is fluid.

Deck drains and weep drain pipes shall not be permanently plugged.

420.39 Installation of Wick Drain – Wick drains shall be installed along the full length of gutters and installed when the asphalt membrane is still hot and tacky. Special attention shall be given to waterproofing and wick drain modifications at deck drains and weep drain pipe locations to ensure proper drainage.

The top surfaces of wick drains must be protected from receiving the tack coat used for asphalt overlay work.

420.40 Application of Protection Board – Protection boards shall be laid on the asphalt membrane:

(a) while the membrane is still hot;

(b) to within 5 mm of all curbs, parapets, wick drains, vertical surfaces of deck drains, and deck joint verticals;

(c) with the length of the board running transversely across the deck;

(d) with edges overlapping a minimum of 12 mm and a maximum of 25 mm both longitudinally and transversely; and

(e) with the longitudinal (direction of traffic flow) joints staggered at least 150 mm.

Protection board shall be rolled by means of a linoleum or lawn type roller while the membrane is still warm, to ensure good contact with the membrane.

Holes shall be cut through the protection board to allow water to drain freely through the wick drains, deck drains and weep drain pipes.

In instances where edges of the protection board curl up, the edges shall be cemented down, using hot membrane material, to the satisfaction of the Ministry Representative.

Protection boards that are warped, distorted or damaged in any way, by manufacture, storage, handling, or exposure to weather, shall be rejected.

420.41 Sampling and Testing – The Ministry Representative may require that sufficient quantities of the asphalt membrane system materials be supplied from the materials being used on the project for immediate analysis, flow tests, water absorption, or for other future testing purposes.

Field testing shall be conducted by the Contractor to ensure membrane thickness meets specifications, and that the membrane, protection board, and asphalt all bond adequately.

420.42 Weep Drain Pipes – Weep drain pipes shall be installed at locations as shown on the Drawings and as per the Special Provisions. Weep drain pipes shall also be installed at all low points on the deck and in the wick drains to ensure there will be no ponding on the membrane.

The Ministry Representative may identify additional locations for weep drain pipes along the edge of the curb or deck to ensure proper drainage of the wick drains and low points. Work to install such extra weep drain pipes will be paid for as Extra Work.

ASPHALT OVERLAY CONSTRUCTION

420.50 Asphalt Paving – Asphalt overlay construction shall proceed only when the final condition of the installed waterproofing membrane system is accepted by the Ministry Representative.

Prior to commencing asphalt pavement construction, the weight of equipment, including loaded trucks, paver and rollers, and their maximum combined weight must be accepted by the Ministry's structural Professional of Record. Only this accepted equipment and equipment combination shall be permitted on the bridge structure at the time of construction.

Asphalt materials, mix design, mix production and hot mix asphalt pavement construction shall meet the requirements of SS 502 and SS 952. Special Provisions will identify any changes to SS 502 bonus/penalty requirements for the bridge application.

420.50.01 Asphalt Mix and Pavement Design – Asphalt mix shall meet the requirements for Pavement Class 1, Medium Mix and appropriate asphalt binder. The maximum aggregate size may be either 16 mm or 19 mm, at the Contractor's option.

The asphalt pavement mix shall be paver laid and compacted in two nominal 50 mm lifts.

NEW SECTION

420.50.02 Placing the Asphalt Mix – In addition to standard paving practice:

(a) the temperature of the asphalt overlay shall meet the membrane manufacturers' recommendations

(b) no asphalt primer shall contaminate the membrane surface;

(c) paving operation shall commence as soon as possible after the waterproofing membrane is installed;

(d) the Contractor shall take due precautions to avoid damage to the membrane system during the paving operations by the paver and hauling trucks;

(e) the paver shall move continuously and at constant speed avoiding any unnecessary stopping or turning;

(f) hauling trucks shall maintain a steady supply of asphalt mix to the paver;

(g) excessive accumulation of asphalt mix in the paver hopper and auger shall be avoided;

(h) unloading of the asphalt mix from the hauling trucks onto the membrane system surface will not be permitted;

(i) paver screed burners shall be set in such a way as to avoid any damage to the membrane system from excessive heat;

(j) the path in front of the screed side end plate must be kept continuously free from spilled asphalt mix. The bottom of the side end plate shall slide freely, otherwise the entrapped asphalt mix could damage the underlying membrane system;

(k) screed adjustments during paving shall be made in such a way as to achieve a specified thickness of mat, surface texture, smoothness and designed pavement geometrics;

(l) the bridge deck joints, drains and other openings shall be protected to avoid intrusion of asphalt during the paving operation; and

(m) during a "one lane at a time" construction method, as on bridges partially open to traffic, the constructed pavement mat shall terminate not less than 150 mm from the edge of installed membrane system in order to provide a sufficient area for lapping of the next portion of the membrane system.

420.50.03 Asphalt Pavement Compaction – In addition to standard compacting practice:

(a) drum vibration mode must not be used during compaction.

(b) steel drum shall run as close as possible to the outer edge, curb or parapet. Outer edge, curb or parapet areas that are inaccessible by the steel drum shall be compacted by a plate compactor.

(c) compaction along the end asphalt plug fillet, expansion joints or concrete dams shall be performed transversely with a steel drum or plate compactor if access by the steel drum is not practicable.

(d) sudden stops or sharp turns by rollers shall be avoided.

420.50.04 Hand Work – Placing of asphalt mix in areas inaccessible to large paving equipment shall be done in accordance with SS 502.

420.50.05 Emulsified Edge Sealant – After completion of paving and compaction, the asphalt roadway surface within 200 mm of a parapet, curb, or other concrete face shall be cleaned of any debris and other materials deleterious to emulsion bonding, and then surface-sealed, using an accepted product under "Asphalt Pavement / Longitudinal Joint Sealing Systems" category of the Ministry's Recognized Products List.

The manufacturer-supplied emulsion may be diluted by the Contractor, at a ratio of 3 emulsion to 1 water.

The diluted emulsion shall be applied at a rate of 0.4 L/m^2 and shall ensure full coverage of the joint along the concrete/asphalt interface.

Blinding sand shall be applied to address any areas of excess application or puddling of the emulsion.

420.50.06 Crack-Control Grooves – Where the asphalt overlay is carried directly over expansion and fixed joints and where indicated on the Drawings, a rectangular groove 20 mm wide and 20 mm deep shall be cut into the overlay. This groove shall be made by either dry sawing or routing, shall have vertical sides, and shall be located directly over the joint for the full length of the joint.

This groove shall be filled with hot-poured asphaltic sealing compound meeting the requirements of the Ontario Provincial Specification OPSS.Prov 1212.

Immediately prior to pouring the sealing compound, the groove shall be dry and cleaned of any dust or debris by an oil-free compressed air jet.

Cakes of joint sealing compound shall be melted on the job site and shall be continuously agitated in the mechanically agitated heating and mixing kettle. The contents shall be continuously agitated until the material can be drawn, free flowing and lump free, from the mixing kettle at a temperature within the range recommended by the manufacturer.

Shields shall be provided to prevent the sealing compound from being spilled on the concrete and on the newly placed bituminous surface.

Sufficient sealing compound shall be poured into the groove so that upon completion of the Work the surface of the compound shall be flush with the surface of the pavement. If the sealing compound subsides to a level below the

NEW SECTION

surface of the pavement, a second pouring shall be done. When more than one pouring is required to fill the groove, succeeding pours shall be made immediately.

MEASUREMENT AND PAYMENT

420.90 Hot Applied Waterproofing Membrane – Payment for Hot Applied Waterproofing Membrane will be made at the Unit Price bid per square metre of deck waterproofed, which price shall be full compensation for the cost of all labour, equipment and materials required for the supply and installation of the waterproofing membrane system including, but not limited to, the preparation of the concrete deck surface, abrasive blasting, supply and application of the primer, asphalt membrane, rubber membrane, membrane and crack reinforcing fabric, wick drains, protection board, and weep drain pipes, handling and controlling of traffic, and everything else done in connection therewith.

420.91 Asphalt Pavement – Payment for Asphalt Pavement will be made at the Unit Price per tonne bid, subject to any EPS payment adjustments per the Special Provisions and SS 502.

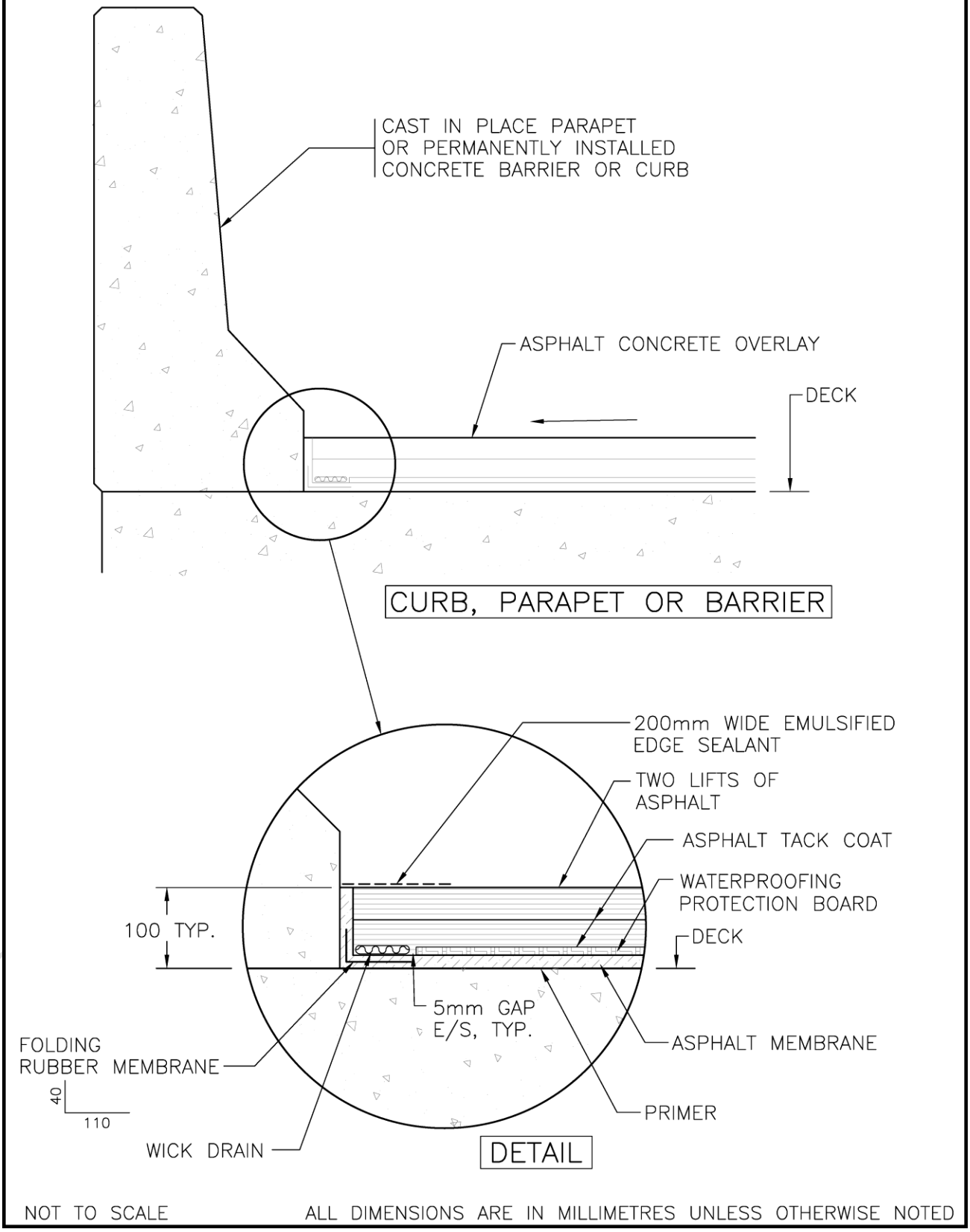
Payment shall include Class 1, Medium Mix pavement mix design, supply and application of asphalt primer and tack coats, asphalt cement and asphalt mix aggregates, quality control, feeding, blending, mixing, loading, weighing, hauling, dumping, spreading, compacting, finishing of the asphalt pavement overlay, cutting and filling crack control grooves in the pavement, supply and application of the emulsified edge primer, quality control, and everything else done in connection therewith.

420.92 Additional Weep Drain Pipes – Construction of any additional weep drain pipes required by the Ministry Representative, beyond the number specified in the Design, will be paid as Extra Work.

NEW SECTION

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE
TYPICAL DETAIL WORK – CURB, PARAPET OR BARRIER

SP420–01

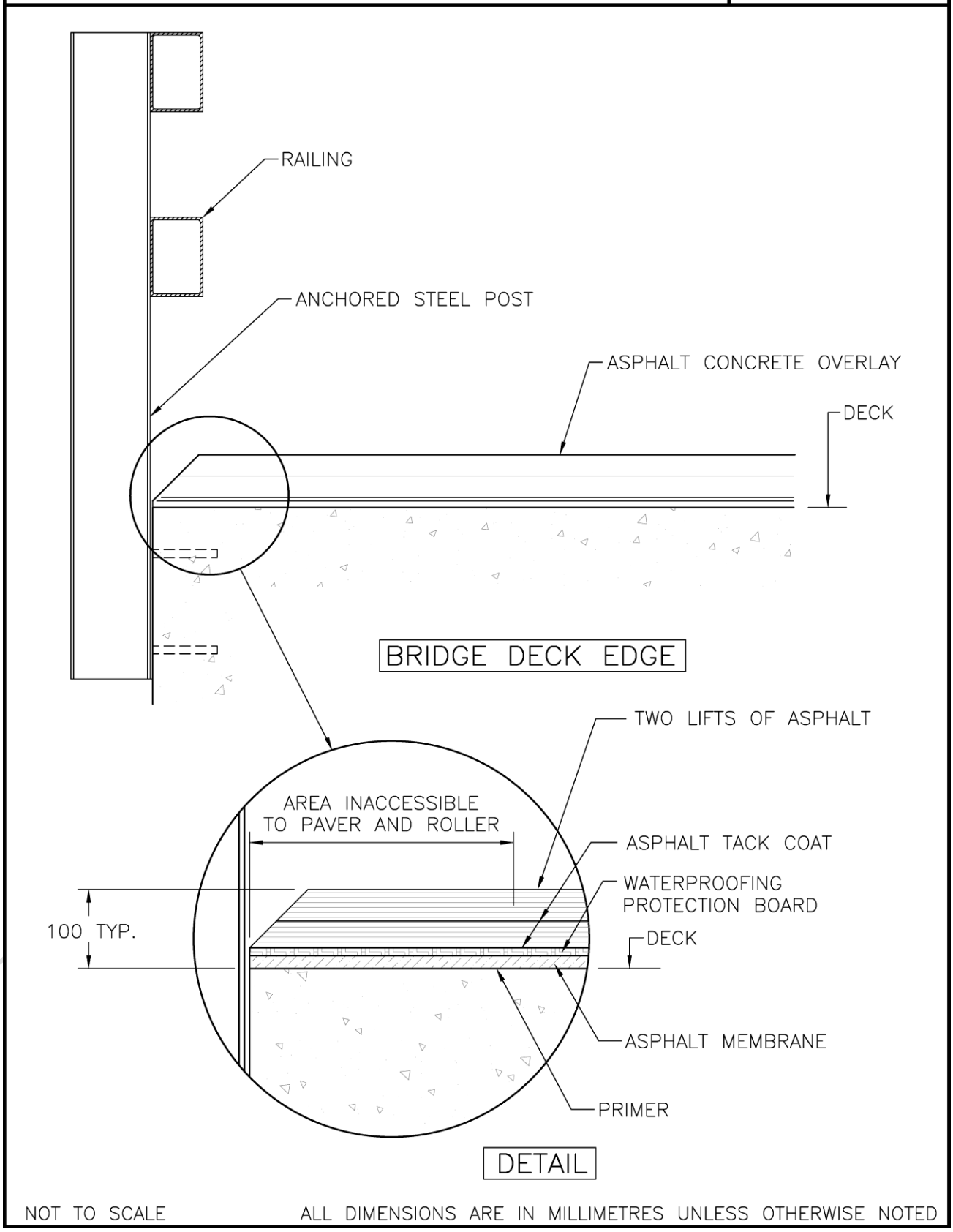


NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE
TYPICAL DETAIL WORK – BRIDGE OPEN DECK EDGE

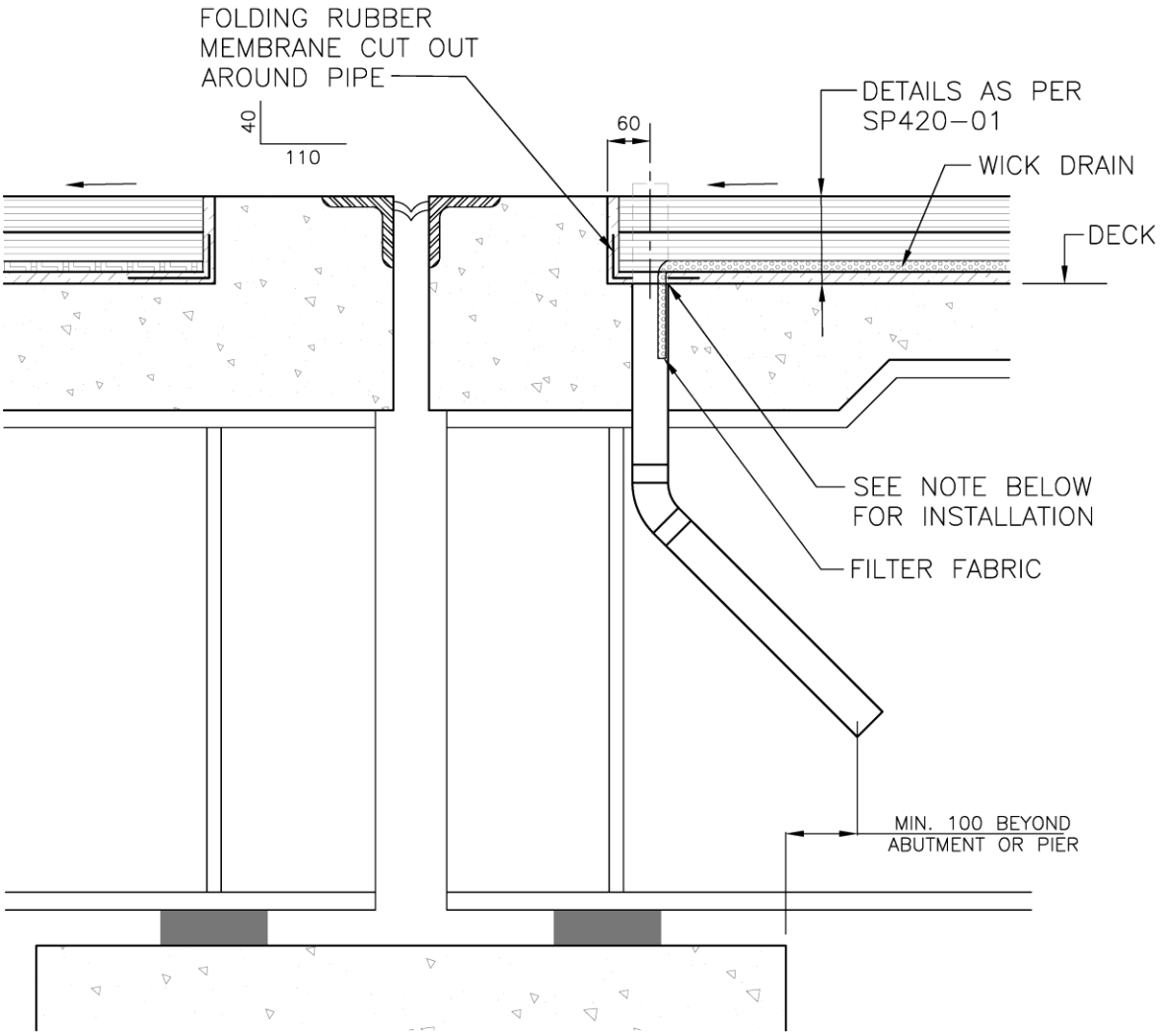
SP420–02



NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE TYPICAL DETAIL WORK – WEEP DRAIN PIPE	SP420–03



WEEP DRAIN PIPE

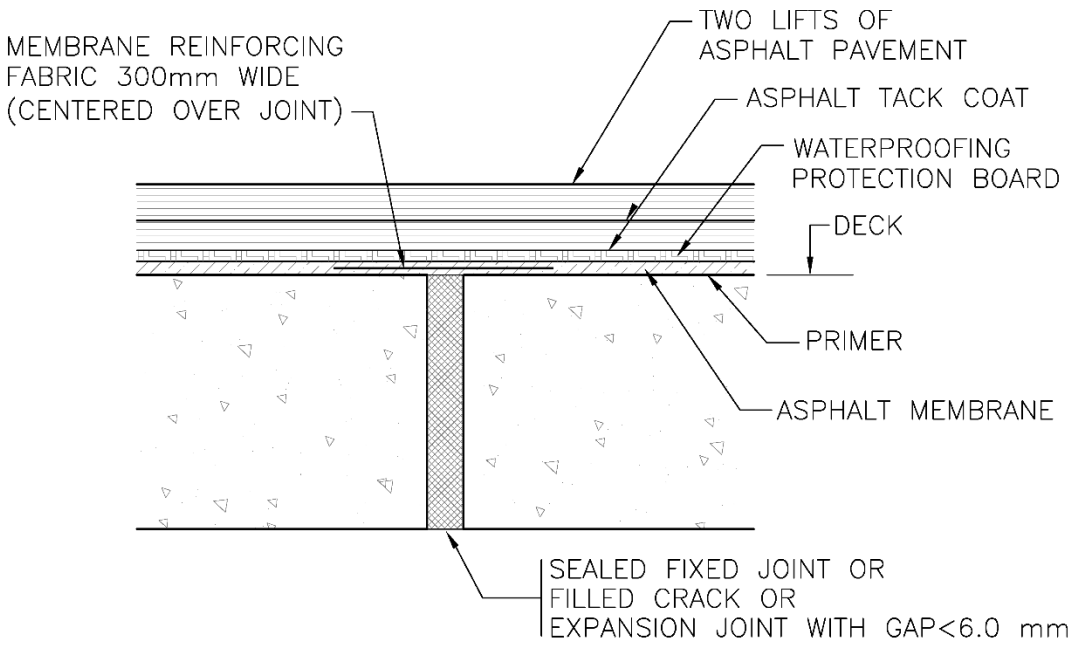
LOCATED AT LOW POINTS

NOTE:
50ø PVC WEEP DRAIN PIPE, TRIM FLUSH WITH TOP OF DECK SURFACE AFTER MEMBRANE HAS CURED. CUT OPEN 100mm OF WICK DRAIN FILTER FABRIC AT ENDS AND TRIM THE PLASTIC INSERT PORTION FLUSH WITH THE INSIDE WALL OF THE DRAIN PIPE. FOLD AND INSERT THE REMAINING FILTER FABRIC INTO PVC PIPE (TYPICAL AT LOW CORNERS OF THE DECK).
DRAIN LOCATIONS SHALL AVOID DISCHARGING ONTO SIDEWALKS, PATHWAYS, TRAFFIC LANES, AND BEARING SEATS.
THIS IS A CONCEPTUAL DETAIL, TO BE REVISED TO SUIT TO STRUCTURE AS NEEDED.

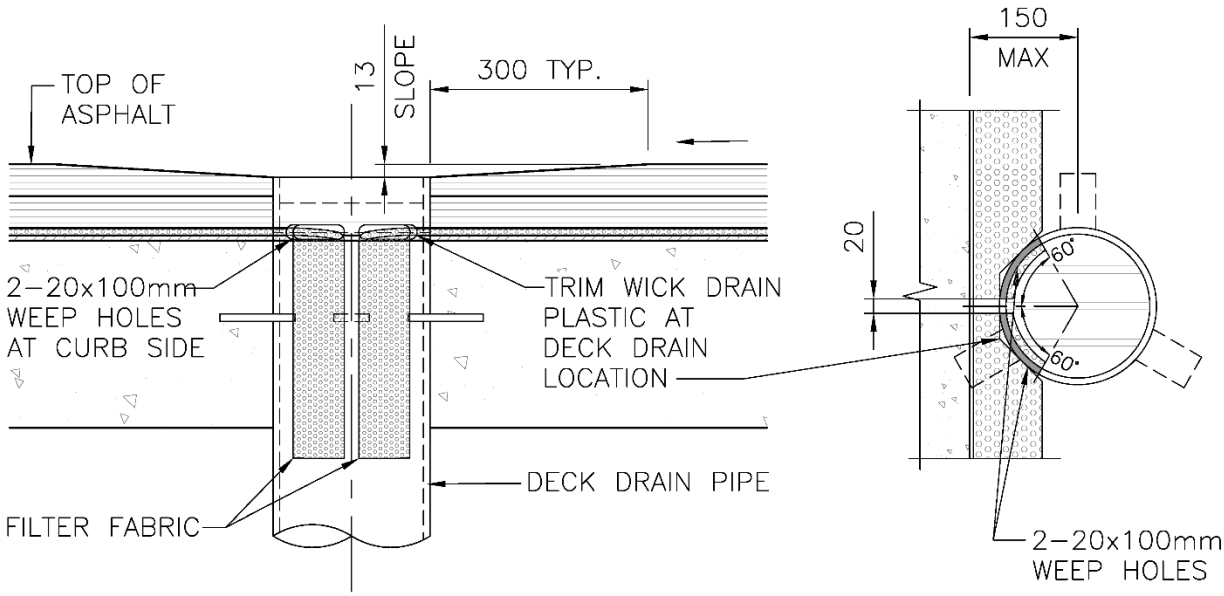
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE
TYPICAL DETAIL WORK – JOINT AND CRACKS, DECK DRAIN

SP420–04



FIXED JOINT, PIER DIAPHRAGM,
CONSTRUCTION JOINT OR CRACK



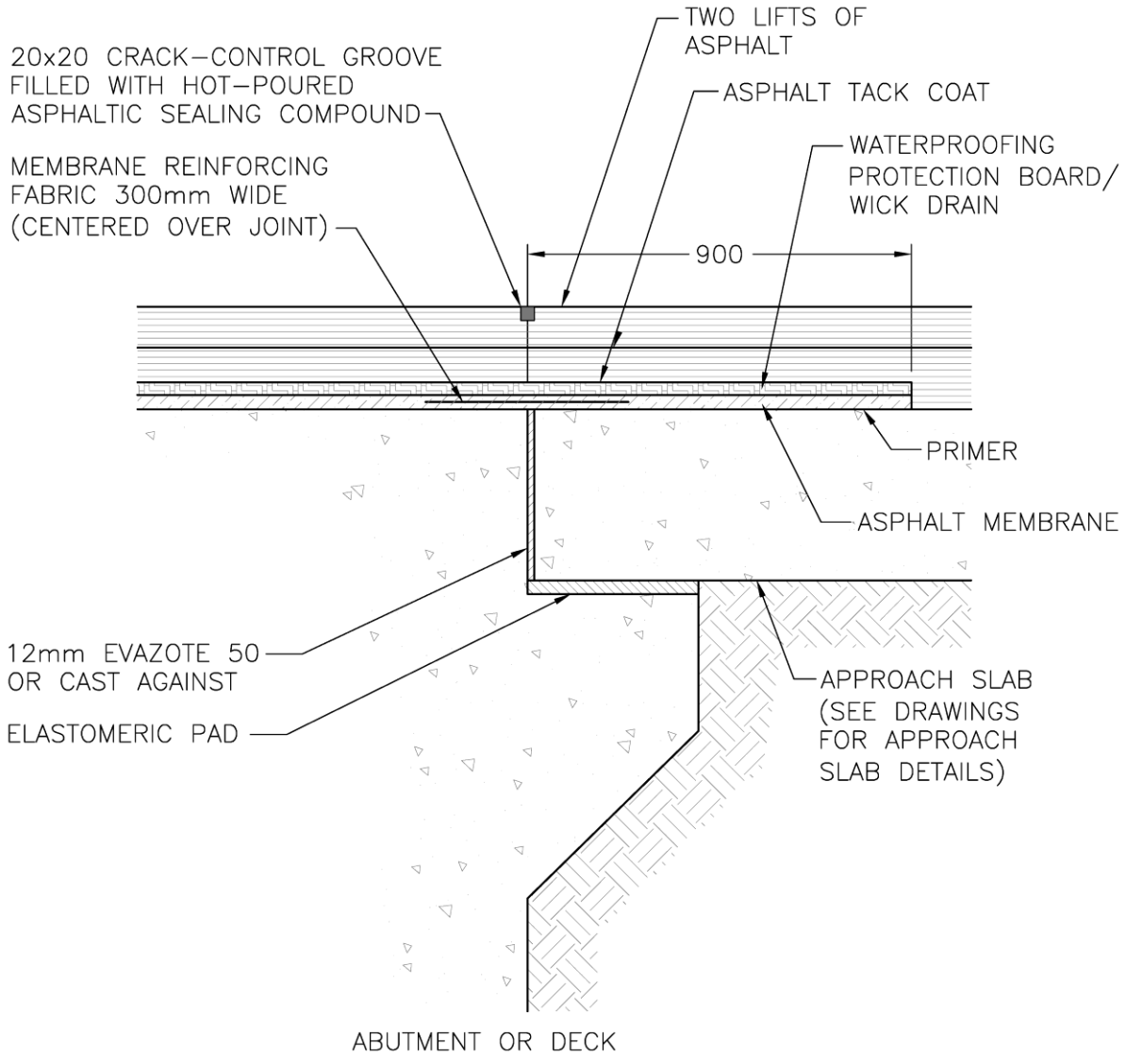
NOTE:
CUT OPEN 300mm OF WICK DRAIN FILTER FABRIC AT EACH DRAIN SLOT AND TRIM PLASTIC PORTION FLUSH WITH THE INSIDE WALL OF THE DECK DRAIN PIPE. FOLD AND INSERT THE REMAINING FILTER FABRIC INTO THE SLOT OF THE DRAIN PIPE.

DECK DRAIN WEEP HOLE DETAIL

NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE TYPICAL DETAIL WORK – APPROACH SLAB	SP420–05



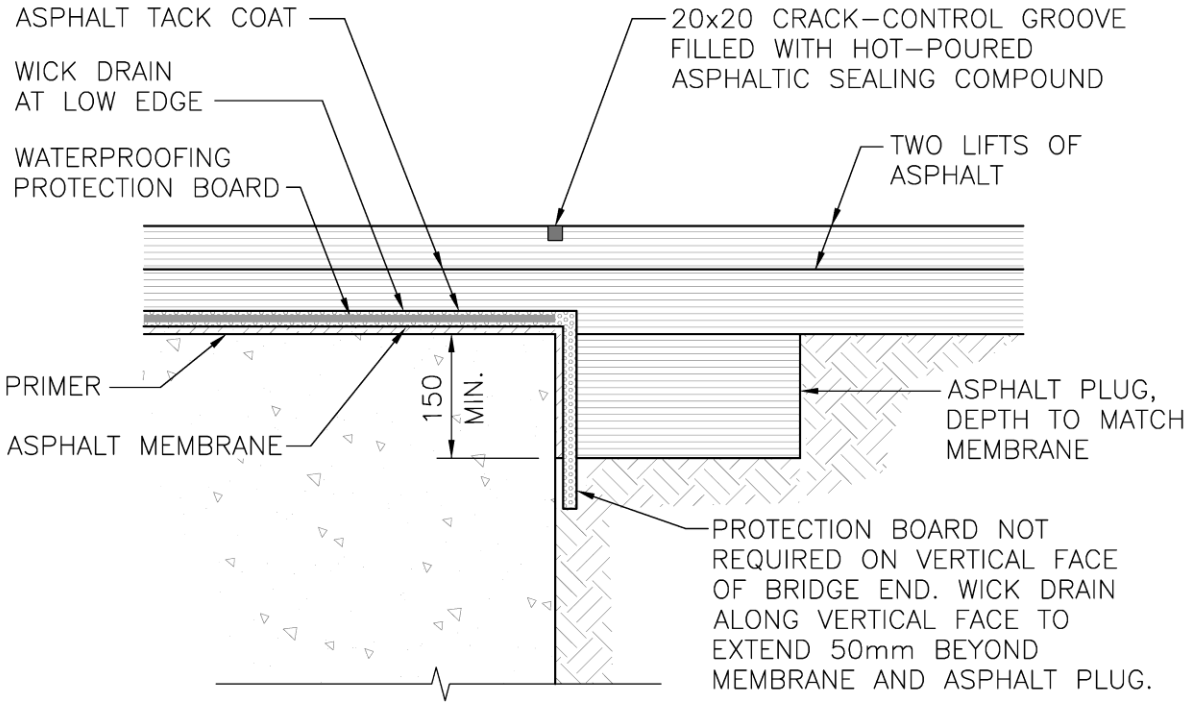
BRIDGE END

(WITH APPROACH SLAB WITHOUT EXPANSION JOINT)

NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE TYPICAL DETAIL WORK – BRIDGE END	SP420–06



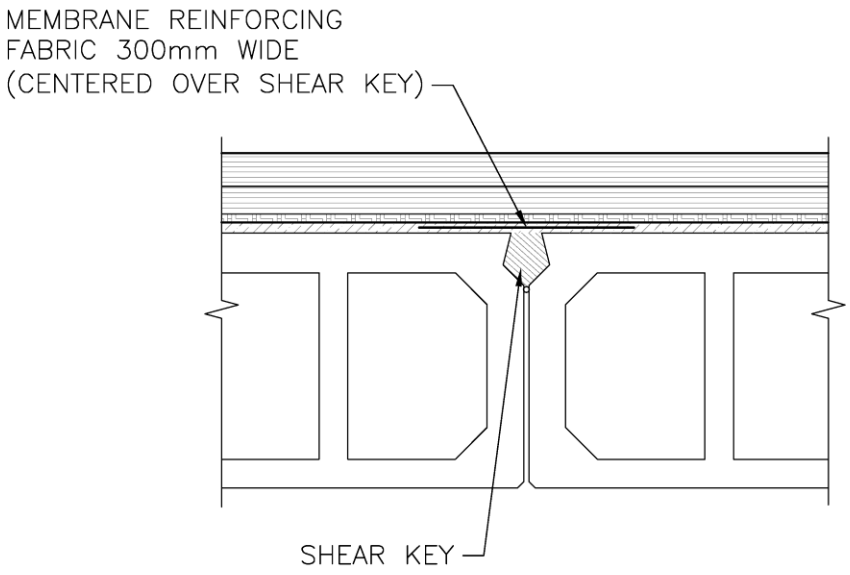
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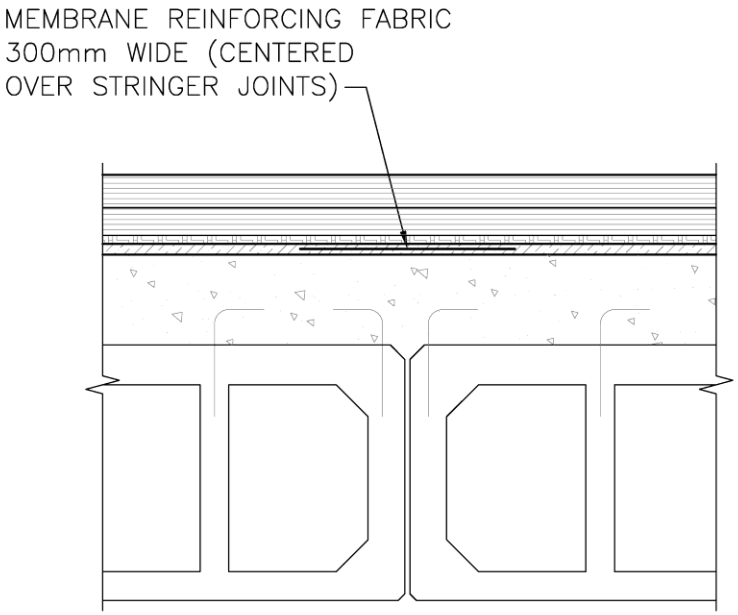
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ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – HOT APPLIED RUBBERIZED MEMBRANE TYPICAL DETAIL WORK – BOX STRINGER JOINT	SP420–07



SHEAR KEY



ABOVE JOINT
WITH COMPOSITE CONCRETE DECK

NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 421
STRUCTURAL STEELWORK

DESCRIPTION

421.01 Scope – This Section covers the supply, fabrication, erection and quality control program of all major structural steel elements.

In general, all shop connections shall be welded, and all field connections high-strength bolted.

Slip-critical bolted connection locations shall be as identified on the Drawings, Special Provisions or as directed by the Ministry Representative.

Except as otherwise specified herein, steelwork shall be fabricated and erected in accordance with CSA S6. Welding and associated Work shall be in accordance with the current edition of CSA W59 – Welded Steel Construction. Fracture control requirements, welding requirements for fracture critical members and welding requirements for primary tension members shall be in accordance with CSA S6 and the Ministry Supplement to CSA S6.

421.02 Prefabrication Meeting – The Ministry will at its discretion, convene a prefabrication meeting with the Contractor to confirm the Ministry's requirements and to review issues such as but not limited to, Schedule, lines of communication, fabricator and sub-fabricator scope of work, location of all work, procedures on quality control, welding procedures, procedures for non-destructive testing, mill certificates and heat numbers, splices, coatings, and any other specific requirements of this specification as it relates to the Work.

421.03 Alternative Details – All details shall, in general, conform to those shown on the Drawings. Any proposed variation shall be submitted in accordance with the provisions of SS 125 – Value Engineering Proposals. The submission of drawings showing alternative details shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia (APEGBC).

Any consideration or implementation of a submitted VEP by the Ministry will also be in accordance with SS 125.

421.04 Working Drawings – Working drawings shall consist of the following:

- Shop drawings
- Transportation details and
- Erection drawings

Transportation details and erection drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia (APEGBC).

When the Contractor is responsible for the design of items that are detailed on the shop drawings, the shop drawings shall be prepared and sealed by a professional engineer registered with APEGBC.

Working drawings shall be in the same system of units as the design drawings.

Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when printed on 11" x 17" sheets.

421.04.01 Shop Drawings

(a) Content – Shop drawings shall show all information and details needed for the fabrication of the members including, but not limited to, such items as:

(i) member shapes, dimensions, and masses,

(ii) camber diagram,

(iii) complete geometric information that member dimensions and shapes are based on,

(iv) material and product standards,

(v) mark numbers and general arrangement of member locations,

(vi) details of connections, attachments, fasteners, and welds,

(vii) special tolerances,

(viii) special handling instructions, and

(ix) coating details, lifting details and lifting locations.

(b) Copies Available at the Fabrication Facility – A copy of the shop drawings shall be available at all times at the location where the components shown on the drawings are being fabricated.

(c) Changes to be Shown on Shop Drawings – Changes to the steelwork from what is shown on the reviewed shop drawings, or repairs made during fabrication and/or construction, shall be indicated by the Contractor on a marked-up set of shop drawings in digital format and submitted to the Ministry Representative at the completion of the Work.

421.04.02 Transportation Details – Transportation details shall include such items as:

(a) Description of hauling and handling equipment

(b) Weight of members

(c) Length and height of loads

(d) Location and method of member support

(e) Details for handling, storing, and loading of members.

421.04.03 Erection Drawings

(a) **Content –** Erection drawings shall be complete in detail of all anticipated phases and conditions during erection and during temporary support of member and shall show in detail the method of erection including, but not limited to, the following:

(i) Erection procedures

(ii) Procedures for off-loading of members upon delivery

(iii) Details for temporary storage and support of members on site prior to erection

(iv) Equipment to be used

(v) Layout or general arrangement drawing showing the layout of the members, equipment positioning, and access roads

(vi) Crane make, model, and capacity charts, boom length(s), crane placement, and access for transporting of members to crane(s)

(vii) Radii and loads for crane lifts

(viii) Rigging details

(ix) Mass of members, rigging and special installation equipment

(x) Details for installation and removal of all falsework, temporary supports, temporary bearings, bracing, guys, dead-men, and lifting devices

(xi) Attachments to the bridge members and bridge structure for temporary support and special launching equipment

(xii) Detailed description of sequence of operations

(xiii) Details for special installation equipment such as a launching truss, launching nose, head frames, spreader beams, and rollers

(xiv) Details for installation of members onto the permanent bearings

(xv) Traffic control plan for roadway and rail traffic

(xvi) Details for protection of existing utilities affected by the erection procedures

(xvii) Layout and details of fall protection and their sequence of installation.

(b) **Lateral Support –** The Contractor shall be responsible for the lateral stability of members at all stages of fabrication, transportation, erection, and subsequent construction, removal, and demolition, and shall design and provide bracing as necessary until completion of the Work.

(c) **Calculations –** Upon request, the Contractor shall submit calculations to the Ministry Representative that demonstrate that specified factored demand/capacity ratios or allowable stresses are not exceeded in members, falsework, temporary bracing and temporary supports and that member capacities and final geometry will be correct. These calculations shall be sealed by the professional engineer who sealed the erection drawings.

(d) Falsework

(i) **Requirements –** Falsework, temporary supports and temporary bracing shall meet the requirements of CSA Standard S269.1, "Falsework and formwork" and shall also meet all the requirements for falsework given in Clauses 20.17 to 20.26 inclusive of the WCB Occupational Health and Safety Regulation.

(ii) **Field Revisions –** A professional engineer registered with APEGBC shall be responsible for any field designs and any changes made to the erection procedures. Field designs and changes to the erection procedures must be documented and sealed by the responsible professional engineer and must be available at the Site prior to the affected erection Work being carried out.

(iii) **Falsework Inspection and Certification –** Immediately before placement of loading on falsework, the Contractor must ensure that the falsework is inspected, and a sealed engineering certificate is issued by a professional engineer registered with APEGBC which:

(A) Indicates the specific areas inspected and

(B) Certifies that the falsework has been erected in accordance with the latest approved erection drawings and supplementary instructions.

421.04.04 Submittals – The Contractor shall submit to the Ministry Representative one set of all working drawings in digital format.

Prior to submission to the Ministry Representative, working drawings shall be reviewed and approved by the Contractor. By this review and approval, the Contractor represents that it has determined and verified all field measurements, field construction criteria, materials, and similar data, and that it has checked and coordinated each working drawing with the requirements of the work and the contract documents. The Contractor shall indicate its review and approval by including on each drawing the date and signature of a person designated by the Contractor as being responsible for the Work.

Working drawings shall be submitted at least fourteen days prior to the fabrication of the Work and shall be accompanied by a transmittal listing each of the drawings submitted. At the time of submission, the Contractor must notify the Ministry Representative in writing of any deviations in the shop

drawings from the requirements of the contract documents. Any Work done or materials ordered prior to the review of the working drawings shall be at the Contractor's risk. The Ministry will review the drawings for general compliance with the contract requirements.

Shop drawings will not be reviewed without the transportation details and erection drawings applicable to the members in question.

Erection will not be allowed to proceed without the Ministry Representative's review of the method proposed.

Review of working drawings shall not relieve the Contractor of responsibility for carrying out the work in accordance with the contract documents.

If so agreed to in advance by the Ministry Representative, working drawings may be submitted in paper format. The reference to working drawing submittal copies shall be increased to four paper copies in this case. At least 14 days before fabrication is to commence, or as otherwise requested by the Ministry Representative, the fabricator shall submit a schedule of fabrication to the Ministry Representative in the form of a Gantt Chart. At the discretion of the Ministry, the schedule shall be updated on no less than a monthly basis. The schedule shall be provided to the Fabricator's Quality Control and to the Ministry Representative for reference and planning of inspections and progress reporting. At the request of the Ministry Representative, the Contractor shall report any interim variations to the schedule.

421.04.05 Working Drawing Format – The Contractor shall transmit working drawings through attachments to e-mail. Unless otherwise agreed to by the Ministry Representative, electronic attachments to an email must total no more than 20 MB and must be submitted unzipped. Drawing files shall be submitted in Portable Document Format (PDF) to print out on 11" x 17" size pages. PDF sets shall be created by "distilling" CAD sheets rather than by scanning paper plan sets. Unless otherwise agreed to by the Ministry Representative, electronic attachments greater than 20 MB in size shall be sent in two parts by separate emails, denoting "1 of 2" and "2 of 2" in the subject lines after other required subject-line information.

If agreed to by the Ministry Representative, the Contractor may employ a document and data management service such as SharePoint® to transmit working drawings. If this process is used, the limitation on drawing file size is waived. The Contractor shall be responsible for setting up the appropriate folders within the document and data management software and for providing access to these folders to the fabricator, Ministry Representative and the design engineer. The Contractor and the Ministry Representative shall send e-mail notification to each other whenever they post drawings to the document and data management folders.

The resolution of drawings shall be such that the finest detail must be legible at full scale on a 21 inch monitor without zooming in (1 in. width on an 11" x 17" sheet is 1 in. on the monitor).

Drawings shall be black images on a white background.

All PDF sheets shall be assembled within a single file, subject to the maximum file size indicated above, ensuring that all sheets are rotated to a "ready to read" orientation within the PDF file set. Generally, 11" x 17" plan sheets should be in landscape and 8½" x 11" note sheets in portrait, so that the majority of text is vertical. PDF sheets shall be ready to print out on appropriately sized paper sheets with no additional formatting required by the viewer.

Drawing sets that are not legible or that do not conform to submission requirements will be returned to the Contractor without being reviewed and the Contractor shall submit four paper copies of the drawing set as substitutes. Drawings of large or complicated pieces, where it is not practical to show all details on an 11" x 17" sheet, may be submitted on full size paper drawings, in which case the Contractor shall submit four copies of the drawing set. When revisions to full size paper drawings are required, the Contractor shall supply four complete sets to the Ministry Representative.

421.05 Quality Control – The Contractor shall implement a quality control program to meet the Contract requirements. The quality control plan shall be made available to the Ministry Representative for review.

421.06 Quality Assurance – The Ministry will implement a quality assurance program by auditing the Contractor's quality control program and by inspection at its discretion.

The Contractor shall notify the Ministry Representative at least 14 days before fabrication has commenced. In addition, the Contractor shall provide a minimum of 48 hours notice to the Ministry Representative that a product will be available for inspection. If the schedule is subsequently changed, the Contractor shall provide the 48 hours notice from the time that the Ministry is notified of this change. If the product is not available or is not sufficiently complete for inspection as notified, at the sole discretion of the Ministry Representative, the Contractor shall be charged stand-by costs for the Ministry's quality assurance inspector. The Contractor shall allow the Ministry representatives safe access to all parts of the Work and shall supply such information and assistance as is required. The Contractor shall provide samples of any materials requested by the Ministry Inspection by the Ministry shall not relieve the Contractor from obligation to perform the Work in accordance with the Contract.

At its discretion and expense, the Ministry will test any weld by non-destructive testing methods, in accordance with CSA W59. Generally, fillet welds will be tested by the dry powder magnetic particle method (MT) and butt joints by radiography (RT), or by any other method of testing the Ministry deems necessary, such as die penetrant (PT), ultrasonic (UT) and phased array ultrasonic testing methods. Selection of the appropriate testing method will be at the discretion of the Ministry.

The Ministry will attempt to schedule non-destructive testing operations so as not to interfere with the progress of the Work.

The extent of non-destructive testing will usually be in accordance with Table 421-A for each girder.

The inspection of welds shall be carried out as soon as possible after the completion of welding, with the exception that restrained joints shall be inspected in accordance with the requirements of CSA W59 Clause 12. The Contractor shall provide reasonable notice to the Ministry Representative/Inspector regarding requirement for inspection of welds.

The Contractor shall be prepared to move and support the pieces being inspected. In general, inspection shall be done on the flat and a minimum of 1.25 m of headroom shall be available.

The Contractor shall bear the cost of re-inspection of welds after defects are repaired. Results of re-inspection shall be provided to the Ministry Representative immediately upon completion of the re-inspection work.

The Ministry will inspect bolted connections in accordance with CSA S6.

The Ministry Representative shall be notified of any defects found in the Work. In general, no repair shall be made until agreed to by the Ministry. In the case of minor corrections, as described by CSA S6-19 Section 10.23.6.4, approval to proceed may be given either verbally by the Ministry's Inspector, or in advance provided that written repair procedures are submitted for Ministry acceptance prior to the Work commencing. In such cases as repair of cracks, or repairs as described by CSA S6-19 Section 10.23.6.5, or a revised design to compensate for deficiencies, the means of correction shall be prepared and sealed by a professional engineer registered with APEGBC. It shall be submitted in writing, with adequate sketches, to the Ministry for review.

The Ministry Representative may reject any items which, in the Ministry Representative's opinion, do not comply with the requirements of the Contract. The Ministry in its sole discretion may back charge all inspection costs for the rejected material to the Contactor.

Each main member or structural component shall be audited by a Ministry quality assurance representative before it is shipped from the shop. This audit shall not relieve the Contractor of responsibility for subsequent damage or for defects which become apparent before the Work is finally accepted by the Ministry. The Contractor shall provide 5 days notice to the Ministry Representative regarding intent to ship a unit or product and the product shall be made available for inspection prior to loading and shipping.

Members shall only be allowed to be shipped out for erection after they have been approved by Quality Control and accepted by the Ministry quality assurance representative. A certificate of compliance may be issued by the Ministry quality assurance representative to acknowledge acceptance.

421.07 Quality and Details of Welds – The quality and details of welds shall be in accordance with CSA W59.

421.08 Design Specifications – CSA S6 and the Ministry Supplement to CSA S6 shall be used in the design of alternative details and calculating the effect of stresses incurred in fabrication and erection.

No increase in allowable stresses due to vertical loads shall be used for erection conditions.

Table 421-A: Non-Destructive Testing for Girders

Element	Test Method	Minimum Extent of NDT
Butt Joints:		
• For joints < 65 mm in thickness	RT	100%
• For joints ≥ 65 mm in thickness	UT	100%
Flange/web fillet welds	MT	100%
Flange/stiffener fillet welds	MT	100%
Web/stiffener fillet welds for ½ depth from Tension Flange	MT	100%
Fillet welds for connection plates for ½ of the depth from Tension Flange	MT	100%
Transverse fillet welds on tension flanges	MT	100%
Other fillet welds	MT	25%
Upon removal of runoff tabs at the edge of a flange member	PT	100%

MATERIALS

421.11 Materials

421.11.01 Structural Steel – Steel shall conform to the requirements of CSA G40.20/G40.21 and shall be of the grades called for on the Drawings. Plates provided from coils shall not be used. Prior to fabrication, the Contractor shall supply to the Ministry Representative the manufacturer's mill certificates giving details of all chemical and physical properties of steel to be used in the Work.

The boron content of steel shall not exceed 0.0008%.

Chemical composition of steel to be galvanized shall be in accordance with SS 421.42.

Steel shall be supplied free of surface defects and internal discontinuities, with due regard for the end use of the steel in the Work.

Edges of all plates will be subject to inspection by the Ministry. Any discontinuities will be examined and may be accepted.

The Ministry Representative shall be supplied with a record of all observed discontinuities and repair methods.

Repairs to defective plates shall not proceed until review of the proposed repair method by the Ministry Representative is completed.

421.11.02 Welding Consumables – All electrodes shall match the base metal specified in accordance with CSA W59, Table 12.1. The deposited weld metal shall provide strength, ductility, impact toughness and corrosion resistance equivalent to the base metal.

Welding consumables for all processes shall be certified by the Canadian Welding Bureau (CWB) as complying with the requirements of CSA W48.

421.11.03 High-Strength Bolts – Unless otherwise specified on the Drawings, high-strength bolts, nuts and washers shall conform to the requirements of ASTM F3125, grade A 325, and as follows:

(a) for painted steelwork, Type 1 bolts shall be provided and painted after installation in accordance with the field painting specifications.

(b) for unpainted portions of weathering steel, Type 3 bolts shall be provided.

(c) for painted portions of weathering steel, Type 3 bolts shall be provided and painted after installation in accordance with the field painting specifications.

(d) for connections of members that are galvanized, Type 1 bolts hot-dipped galvanized to the requirements of ASTM A153/A153M shall be provided.

421.11.04 Shear Connector Studs – Shear connector studs shall conform to the requirements of ASTM A108, Grades 1015, 1018, or 1020.

421.12 Material Storage and Care

421.12.01 Steel – Structural material, either plain or fabricated, shall be stored above the ground in an upright and shored position upon platforms, skids, or other supports unless otherwise permitted by the Ministry. Should permission be granted to stack the steel in a horizontal position, the Contractor shall provide sufficient support of the steel in order to prevent bending of the plate prior to incorporation into the Work. It shall be kept free from dirt and other foreign matter and shall be protected as far as practical from corrosion. Long members shall be supported on skids placed near enough together to prevent overstress from deflection.

Prior to fabrication, all steel shall be marked for identification by heat number and specification by a marking system approved by the Ministry.

421.12.02 Welding Consumables – Electrodes and fluxes shall be stored and kept in condition as required by CSA W59, Section 5.2.

Gas for welding shall be stored in marked steel bottles and shall not be subjected to temperatures in excess of 50°C or temperatures of less than 0°C.

421.13 Fabrication – Prior to fabrication, the welders and welding operator's qualifications, shop drawings, welding procedures, mill certificates and welding consumable certificates shall be submitted for the Ministry's review.

No fabrication, welding or coating of steelwork shall commence until permission to do so has been received from the Ministry.

421.14 Ministry's Plant Office – The Contractor shall provide in the fabrication plant or nearby a suitable lock-up office for the sole use of the Ministry Representative throughout the period of fabrication.

The office shall be at least 2.5 m x 3 m and shall be weathertight and reasonably soundproof, provided with ample window area and ventilation, furnished with a 0.75 m x 1.2 m desk, a 1 m x 2 m drafting table, a two-drawer filing cabinet, two chairs, a drafting stool and 3 m of shelves, and equipped with a telephone, and high-speed internet access which shall be operational throughout the period of fabrication. All long-distance telephone calls will be paid for by the Ministry.

The Contractor shall clean, heat, and light the office throughout the period of fabrication.

EQUIPMENT

421.21 Qualifications and Equipment

421.21.01 Contractor – The Contractor shall produce evidence that the plant is currently fully certified by the CWB to the requirements of CSA W47.1, Division 1 or Division 2 prior to beginning Work.

The Fabricator shall also produce evidence of at least three years of satisfactory experience in the fabrication of bridge structural steelwork.

The Contractor shall employ or retain a registered professional engineer experienced in steel bridge fabrication, to provide guidance throughout the work.

A qualified welding supervisor shall be employed on each shift where welding is done on the work.

Prior to fabrication, the Contractor shall submit to the Ministry the names of the welding engineer, the engineer experienced in steel bridge fabrication, and the welding supervisors who are to be employed on the Work.

421.21.02 Certification – Structural steel elements shall be fabricated in plants that have third party certification under an industry recognized quality certification program specific to steel bridge fabrication.

Acceptable third party certifications include:

(a) the Canadian Institute of Steel Construction Steel Bridge Certification – Complex Bridges or

(b) the American Institute of Steel Construction Certification Program for Steel Bridge Fabricators, in either the intermediate or advanced bridge categories, as applicable to the type of Work.

Certification shall be in effect prior to the beginning of the Work and shall be maintained throughout the period of manufacture.

421.21.03 Welders and Welding Operators – The Contractor shall produce evidence that all welders and welding operators to be employed on the Work are currently qualified by the CWB at the time of fabrication in the processes in which they are to be employed on the Work.

The Contractor shall also produce evidence relative to each welder and welding operator, that they have been executing satisfactory welding in the required processes within the six-month period previous to the award of this Contract.

421.21.04 Welding Equipment – All equipment to be used in the Work shall be in good working order and shall be subject to the inspection of the Ministry.

For any arrangement of automatic welding, the Ministry Representative may require that a preliminary test run of the equipment be made, without welding, over the length of the joint, to prove that the disposition of the equipment and the method and accuracy of travel are satisfactory.

421.22 Welding Procedures – The Fabricator shall submit copies of the proposed welding procedures for review by the Ministry. Gas metal arc welding shall not be used. Where the submerged arc or flux cored arc process is to be used, the Ministry Representative may order that each welder and welding operator make a weld specimen not less than 1 m in length for fillet welds and 150 mm in length for butt joints. Steel of the same specifications and thickness as that to be used in the work shall be used in the specimen welds.

No welding shall be done on the Work until the welders' and welding operators' qualifications are established to the satisfaction of the Ministry Representative.

Welding procedures shall be accompanied by documentary proof that they have been qualified previously by the CWB at the plant where the Work is to be carried out.

The procedures shall include the following information: joint type, welding process, welding position, base metal specification, welding consumable specification and size, preheat requirements, amperage and voltage requirements, speed, polarity, and welding equipment, including a description of travel for automatic welding. Additional information, as described in CSA W47.1, Annex C, shall also be included in the procedures.

421.23 Butt Joints – Except as called for on the Drawings, or as reviewed and approved by the Ministry's design engineer, butt joints will not be permitted. Approval must be given prior to fabrication.

The Fabricator may submit an alternative butt joint design, to that shown on the Drawings, or propose a butt joint design if one is not shown on the Drawings, provided that all such designs have been approved by the CWB. Acceptance of alternative butt joint designs shall be subject to approval by the Ministry. Location and details of all butt joints shall be explicitly shown on the shop drawings.

421.24 Assembly and Welding Sequences – If requested by the Ministry Representative, the Contractor shall supply full details of the proposed assembly and welding sequence of any particular weldment.

CONSTRUCTION

421.31 Preparation of Material

421.31.01 Straightening Material – Prior to being used in fabrication, all structural steel shall be straight and free from kinks or bends. The flatness tolerance of plate in excess of 900 mm wide shall be in accordance with the tolerances of the finished product as specified in CSA W59. If straightening is necessary, it shall be done by methods that will not injure the metal. The steel shall not be heated unless permission is given by the Ministry Representative.

In no case shall the temperature of the steel exceed 620°C. After straightening, the surfaces of the metal shall be carefully inspected for evidence of fracture and if necessary, the material shall be replaced or repaired to the satisfaction of the Ministry Representative. Sharp kinks and bends will be cause for rejection of the steel.

421.31.02 Camber – Girders shall be cambered as shown on the Drawings.

421.31.03 Edge Preparation – Steel may be cut to size by sawing, shearing, plasma cutting or flame cutting. All steel before cutting shall be marked by a method agreed to by the Quality Manager and Ministry Representative so that its specification may be immediately identified.

All cut edges shall be smooth and regular, free from fins, cracks, tears and notches. Freehand cutting shall be done only when and where approved by the Ministry Representative. Roughness of cut surfaces shall not exceed the American National Standards Institute ASME B46.1 value of 1000. Roughness exceeding this value shall be removed by machining or grinding. Occasional gouges will be tolerated only at the discretion of the Ministry Representative and shall be repaired in accordance with a procedure approved by the Ministry.

Sheared edges of plates more than 16 mm in thickness shall be planed to a depth of 6 mm.

Re-entrant flame cuts shall be cut to a radius of not less than 20 mm.

All exposed edges and corners of members, rolled or cut, which are to be coated, shall be ground to a minimum radius of 2 mm.

All corners of oxygen cut and plasma cut edges of main stress-carrying members including the outside edges of flange splice plates, except unpainted bearing stiffeners and girder webs, shall be ground to a minimum radius of 2 mm.

Special attention shall be given to the cutting of flange plates. All flange plates prepared by flame cutting shall be preheated in accordance with SS 421.34.

The Fabricator will carry out Brinell hardness testing of the edges of flange plates and flange splice plates on fracture critical members. For flange plates, readings shall be taken near both ends and the centre on each side of the plate for a total of 6 readings minimum per plate. For flange splice plates readings shall be taken near the centre on each side of the plates for a total of two readings minimum per plate. The locations and readings shall be recorded for review by the Ministry Representative. The Brinell hardness of the edges of flange plates and flange splice plates on fracture critical members shall not exceed 220. If greater hardness is measured, the edges shall be ground by the Fabricator to remove the hard layer. The Ministry Representative may conduct Quality Assurance testing of Brinell hardness once the flanges have been prepared to meet the hardness requirements. The Contractor shall bear all costs for re-inspection and retesting.

Surfaces to be welded shall be free from loose scale, slag, rust, grease, moisture or other material that will prevent proper welding. Mill scale that withstands vigorous wire brushing, a light film of drying oil or a thin rust inhibitive coating may remain except that all mill scale shall be removed from the surfaces on which flange-to-web welds are to be made. Surfaces within 100 mm of any weld location shall be free from any paint or other material that would prevent proper welding or produce objectionable fumes while welding.

Edges of material thicker than specified in the following list shall be trimmed if and as required to produce a satisfactory welding edge wherever a weld along the edge is to carry calculated stress:

(a) Sheared edges of material thicker than 12 mm;

(b) Rolled edges of plates (other than Universal Mill Plates) thicker than 10 mm;

(c) Toes of angles or rolled shapes (other than wide flange sections) thicker than 16 mm;

(d) Universal Mill Plates or edges of flanges of wide flange section thicker than 25 mm.

421.31.04 Direction of Rolling – Steel plates for main members and splice plates for flanges and main tension members shall be cut and fabricated so that the direction of rolling is parallel to the direction of the primary stresses.

421.31.05 Bolt Holes – Standard holes for high tensile bolts shall be either punched, sub-punched and reamed, or drilled, and shall be of a nominal diameter not more than 2 mm in excess of the nominal bolt diameter, except that the following bolt/hole combinations will be permitted:

(a) either 3/4 inch or M20 bolts in 22 mm holes;

(b) either 7/8 inch or M22 bolts in 24 mm holes;

(c) either 1 inch or M24 bolts in 27 mm holes.

Oversize or slotted bolt holes will be allowed only in special circumstances or as shown on the Drawings.

Punched holes shall be clean cut, without torn or ragged edges. The diameter of the die shall not exceed the diameter of the punch by more than 2 mm. If a punched hole must be enlarged to admit a high tensile bolt, it shall be reamed.

Reamed holes shall be cylindrical and perpendicular to the member. Where practicable, reamers shall be directed by mechanical means. Reaming shall be done with twist drills or reamers.

Drilling shall be done with twist drills or core drills. Burrs on the outside surfaces shall be removed.

Poor matching of holes will be cause for rejection.

421.31.06 Faying Surfaces – In the parts of a structure where steelwork is specified to be painted, faying surfaces shall meet the requirements of SS 216.

In the parts of a structure where steelwork is specified to be uncoated, the following shall apply:

(a) For Non-Slip Critical Uncoated Connections – faying surfaces shall be free from loose scale, unacceptable burrs, dirt, oil and foreign material.

(b) For Slip-Critical Uncoated Connections

(i) Class A – Faying surfaces shall have a clean uncoated mill scale surface free from loose scale, paint, lacquer, or any other coating in all areas within the bolt pattern and for a distance beyond the edge of the bolt hole that is the greater of 25 mm or the bolt diameter.

(ii) Class B – Faying surfaces shall be blast cleaned to SSPC-SP10 / NACE No. 2.

When assembled, all joint surfaces, including those adjacent to bolt heads, nuts, and washers, shall be free from loose scale, unacceptable burrs, oil, dirt, and foreign material that would prevent the solid seating of the parts.

421.32 Marking – Prior to fabrication, all steel shall be marked for identification by heat number and specification by

a marking system approved by the Ministry Representative. Steel which is unidentified shall not be used in the Work.

421.33 Assembly of Weldments – The shop assembly of the various components of the weldments shall be executed in accordance with CSA W59.

Tack welding shall be done by qualified welders, using the smallest size weld required to hold the components of the assembly together. Tack welds shall be incorporated into the final weld.

421.34 Preheat and Interpass Temperatures – No welding shall be done when the temperature of the base metal is lower than -18ºC. At temperatures below 0ºC, the steel shall be preheated to a temperature of at least 10ºC in excess of that stated in CSA W59, Table 5.3.

Preheat shall be applied to all steel to be welded so that the steel within the greater of 75 mm of the weld, or the thickness of the thickest part to be welded, is heated to the temperatures shown in CSA W59, Table 5.3.

Preheat shall be applied in such a manner that moisture from the heating equipment does not penetrate the joint.

Preheat temperatures above the minimum shown in CSA W59, Table 5.3 may be required for highly restrained joints if designated by the Ministry.

Preheat and interpass temperatures for repair of fracture critical members shall be in accordance with CSA S6 Table 10.16.

Preheat temperature shall in no case exceed 200ºC.

Preheat requirements for tack welds shall be meet the above requirements except that where single pass tack welds are used and are to be incorporated and consumed in a weld made by the submerged arc process, preheat is unnecessary.

421.35 Welding – Welding shall be done by the shielded metal arc, metal-cored arc, flux-cored arc or submerged arc processes in accordance with the reviewed procedures and CSA W59, Section 5.

Any weld between the web and flange of a bending member shall be made by a mechanized submerged arc process, which provides a continuous weld throughout the length of the member.

All welding shall be done under cover and, in the case of flux-cored arc welding, shall be done in an area free from wind or draft.

Where the submerged arc process is to be used, the Contractor shall:

(a) Carry out a preliminary test run of the procedure over the length of the joint to prove that the disposition of the equipment, the handling of hoses, and the method and accuracy of travel are satisfactory.

(b) Have each operator make a weld specimen not less than 1200 mm in length for fillet welds and 150 mm in length for butt welds. Steel of the same specification and thickness as that to be used in the Work shall be used in the specimen welds. No welding shall be done on the Work until such a specimen is satisfactory to the Ministry Representative.

Welds in butt joints shall be extended beyond the edges of the parts to be joined by means of start and run-off tabs providing sufficient thickness to avoid the weld burning through and with a joint preparation similar to that on the main material. Welds that connect run-off tabs to the main member shall be placed inside the joint so that they are incorporated into the final weld. For manual shielded metal arc welding the width of the tabs shall be not less than the thickness of the thicker part being joined or 75 mm, whichever is greater. For submerged arc, welding the width of the tabs shall be not less than 75 mm. Each weld pass shall be carried far enough beyond the edge of the parts being joined to ensure sound welds in the joint. Tabs shall be removed upon completion by flame cutting and grinding. The weld shall be cooled without damage to the parent plate. The end of the weld shall be made smooth and flush with the edges of the abutting parts.

In flux-cored arc welding the equipment shall be capable of sustaining a gas flow rate of from 0.85 to 1.25 m^3/h.

Where, in the opinion of the Ministry Representative, excessive repairs are required during the fabrication of components, the fabricator shall submit a plan to the Ministry Representative for review and acceptance, which shows revisions to the fabrication process, including personnel if necessary, to reduce or eliminate future fabrication non-conformances.

421.36 Shear Connector Studs – Shear connector studs shall be welded in the locations shown on the Drawings, to the requirements of CSA W59, Section 6. Shear connectors will be inspected and shall be repaired if necessary in accordance with the same standard.

421.37 High Strength Bolts – Installation of high-strength bolts shall be in accordance with the CSA S-6 "turn of nut" method.

Galvanized bolts shall be lubricated with beeswax or other approved lubricant before installation.

Heads of bolts shall be placed on the outsides of girders and box members, and generally on the more conspicuous side, if any, of any connection.

421.38 Bent Plates – When bending plates, the plates shall be so taken from the stock plates that the bend line will be at right angles to the direction of rolling. Before bending, the corners of the plate shall be rounded to a radius of 2 mm throughout that portion of the plate at which bending is to occur. Bending shall be done by methods that will not crack, tear or otherwise injure the metal.

Cold bending and the minimum radii for cold bending shall be in accordance with CSA S6.

Hot bending of plates shall not be permitted without prior approval of the Ministry. Any proposed method for hot bending of plates shall be submitted to the Ministry for review and approval by the Ministry prior to proceeding with the work.

Bending of flange plates shall not be permitted without prior approval of the Ministry. Any proposed method for bending flange plates shall be case specific and shall be submitted to the Ministry for review and approval by the Ministry prior to proceeding with the work.

421.39 Shop Assembly of Bolted Connections – Holes in girder and truss field splices shall be drilled while assembled in the shop, or subpunched or subdrilled and reamed while assembled. Unless otherwise specified, the structure shall be progressively assembled in accordance with CSA S6. The Ministry shall be provided sufficient notice by the Fabricator, to allow inspection of any assembly, including camber, alignment and accuracy of holes before drilling or reaming is commenced.

Connecting parts assembled in the shop for the purpose of reaming or drilling holes shall be match-marked by a method agreed to by the Ministry to indicate the location and orientation of all pieces.

Alternatively, the Contractor may drill holes full-size using automatic drilling equipment. In this case a check assembly will be required for the first of each major structural type. Reaming of holes shall be as specified in SS 421.47, if the bolt holes do not line up during field assembly.

Progressive assembly in the yard may be allowed, subject to approval by the Ministry Representative, provided the Fabricator can demonstrate that yard assembly will provide as high a quality product as shop assembly.

421.40 421.40 Dimensional Tolerances – Except as noted herein, the dimensions of completed members shall comply with the appropriate dimensional tolerances as specified in CSA W59.

Alignment or position of secondary members shall be within ±6 mm.

The tolerance in flange width shall be ±(b/100), where b is the flange width, but not less than 5 mm and not greater than 25 mm.

The tolerance on the width of stiffeners and plates for secondary members shall be -3 mm to +10mm.

Misalignment of stiffeners on opposite faces of a web shall be less than one third of the web thickness for bearing stiffeners and half the web thickness for intermediate stiffeners.

The maximum deviation from specified length shall be ±L/1000 but not over 20 mm.

Warpage of box members shall be determined by taking measurements at any two cross-sections in a member, at opposite edges of one face of the member. Warpage is defined as the distance by which any point deviates from a plane defined by the other three points. This warpage shall not exceed 1/200 of the width of the member, or 3 mm, whichever is greater.

421.41 Machined Surfaces – Machine-finished surfaces, as designated on the Drawings, shall be coated with an approved protective compound.

421.42 Galvanizing and Metallizing

421.42.01 Galvanizing – All steelwork to be galvanized shall be identified in the Drawings, Special Provisions or elsewhere in the Contract.

(a) Specifications – All steelwork to be galvanized shall be galvanized after complete fabrication to the requirements of ASTM A123M and ASTM A385.

(b) Embrittlement – The galvanizer shall safeguard against embrittlement as required in ASTM A143. Galvanized members shall be subject, at the discretion of the Ministry Representative, to the tests for embrittlement outlined in ASTM A143.

(c) Steel Composition – The chemical composition of steel being galvanized shall be as follows:

- **(i)** Carbon less than 0.25%
- **(ii)** Phosphorus less than 0.04%
- **(iii)** Manganese less than 1.3%
- **(iv)** Silicon less than 0.04% or between 0.15% and 0.22%

(d) Special Techniques – For steel not meeting these chemical composition requirements, special galvanizing techniques shall be developed by the galvanizer to ensure that the specified coating thickness and adherence is achieved. A detailed description of the special techniques shall be submitted to the Ministry Representative for review two (2) weeks prior to galvanizing.

421.42.02 Metalizing – All steelwork to be metallized shall be identified in the Drawings, Special Provisions or elsewhere in the Contract.

(a) Specifications – All steelwork to be metallized shall, after complete fabrication, be treated in accordance with the current SSPC-CS23.00 / AWS C2.23M / NACE No. 12.

(b) Coating Thickness – The zinc coating shall not be less than 0.3 mm in thickness.

421.43 Shop Painting – The Drawings or Special Provisions shall specify whether the structure is to be painted, or what parts of a structure are to be painted. This subsection applies to those parts of the steelwork which are to be painted.

If painting is specified on the Drawings or in the Special Provisions, the structural steel of weathering steel bridges

shall be painted for the larger of the following two distances from deck joint locations:

- 3000 mm; or
- 1.5 times the superstructure depth (including girder, haunch and slab thickness).

Coating work shall be in accordance with SS 216 unless noted otherwise in this SS 421. For the short lengths of coating required adjacent to deck joints on weathering steel bridges, third party Quality Control inspectors shall be qualified to at least AAMP CIP Level 2.

All coatings shall be applied according to their manufacturer's product data sheet unless specified otherwise in the Contract Special Provisions.

All steelwork which is to be painted, except as noted herein, shall be given three shop coats of paint – primer, stripe coat and midcoat. Paint shall be chosen from SS 308 System SS1. Paint shall be supplied by the Contractor.

The topcoat may be applied in the shop or in the field. The topcoat coat shall be chosen from either SS 308 System SS1 or SF2. If the topcoat is to be applied in the shop, then the Quality Control Program shall specifically address the integrity of the topcoat through to project completion.

Surfaces surrounding bolt holes at connection locations shall receive the prime coat only and shall be masked off so that no stripe coat, midcoat or topcoat paint will be under the bolt heads, washers or nuts. These masked off areas shall have the stripe coat, midcoat and topcoat paint applied in the field after installation of the bolts.

Faying surfaces shall receive the prime coat only, as per SS 216.12.01(b).

Paint shall be applied in a covered area in accordance with the manufacturer's specifications and SS 216.

All edges, corners, crevices, bolts, nuts, protrusions and welds (unless ground flush) shall be stripe painted by brush for a width of 50 mm with the midcoat paint before the midcoat coat is applied. Stripe coating paint shall be applied as per SSPC-PA1 and may be applied by spray, but shall be brushed in. Stripe painting shall be allowed to cure before the midcoat coat is applied.

Unless a different coating is called for on the Drawings, the exposed steel surfaces of bearings, and bearing assemblies, shall be cleaned and painted as structural steelwork.

Surfaces shall be cleaned to "near-white" per SSPC-SP10 / NACE No. 2 and 50 to 75 μm (2 to 3 mils) sharp/angular profile.

The midcoat shall not be applied until the primer and stripe coat are accepted by the Ministry Representative to be sufficiently cured.

Any surfaces inaccessible after erection, except faying surfaces and tops and sides of top flanges, shall be given the topcoat in the shop.

If the topcoat is to be applied in the shop, then the topcoat shall be applied after the midcoat is accepted by the Ministry Representative to be sufficiently cured.

The colour of the topcoat will be selected by the Ministry. For weathering steel, unless noted otherwise, the topcoat colour shall match the expected colour of the oxidized surfaces (Federal Color 30045 unless noted otherwise). The proposed colour shall be subject to the acceptance of the Ministry Representative.

421.44 Marking and Shipping – Each member shall be marked by a method agreed to by the Ministry Representative with an erection mark, corresponding to the mark shown on the erection diagram.

Members shall be loaded in such a manner that they can be transported to and unloaded at their destination without being damaged.

After steelwork has been delivered to Site it shall be inspected by the Contractor's quality control inspector. The Contractor shall clean the steelwork after it has arrived at Site of any dirt, road salts, slush or other contaminants accumulated during transport and shall carry out any other surface preparation work necessary to meet the specified surface preparation requirements.

421.45 Field Assembly – The parts shall be accurately assembled as shown on the drawings and any matchmarks shall be followed. Hammering which will injure or distort the members shall not be done. Bearing surfaces and surfaces to be in permanent contact shall be cleaned before the members are assembled. Field connections shall have one half of the holes filled with bolts and cylindrical erection pins (half bolts and half pins) before final bolting. Fitting-up bolts shall be the same nominal diameter as the high tensile bolts, and cylindrical erection pins shall be 1 mm larger.

421.46 Straightening Bent Material During and after Fabrication – The straightening of plates and angles or other shapes shall be done by methods that will not produce fracture or other injury. Any proposed straightening plan, whether by heating and or mechanical straightening methods shall be case specific and shall be submitted to the Ministry Representative for review and acceptance prior to commencing with the work. Generic plans may be referred to as part of the straightening plan. The heating shall not exceed the requirements of CSA W59, Clause 5.15. After heating, the metal shall be cooled as slowly as possible, typically at ambient shop temperature.

Following the straightening of a bend or buckle, the surface of the metal shall be carefully inspected for evidence of fracture, and if necessary, replaced or repaired to the satisfaction of the Ministry. The cost of any non-visual inspection deemed necessary by the Ministry Representative